

Research Interests

My current research focuses on 3 areas of Mechanistic Interpretability for Large Language Models (LLMs):

- **Activation Space Geometry (1):** Understanding how concepts and behaviors are represented in the activation space of LLMs. To what extent the Linear Representation Hypothesis and the Platonic Representation Hypothesis hold? [1]
- **Application of Mechanistic Interpretability (2):** Activation Steering, Model Editing, Model Fingerprinting, and other applications. [1, 2]
- **Modular Decomposition (3):** Decomposing LLMs into smaller, interpretable modules that perform specific functions, and studying their interactions. [1, 4, 3]

Publications

*: co-first authorship

- [1] **Hieu M Vu** and Tan Minh Nguyen. “Angular Steering: Behavior Control via Rotation in Activation Space”. In: *NeurIPS* (2025). **Spotlight (top 3.5%)**. Poster. ArXiv. Github.
- [2] Dung V Nguyen*, **Hieu M Vu***, Nhi Y Pham*, Lei Zhang, and Tan M Nguyen. “Activation Steering with a Feedback Controller”. In: *ICLR* (2026). ArXiv. Github.
- [3] Tuan-Anh D Nguyen, **Hieu M Vu**, Nguyen Hong Son, and Minh-Tien Nguyen. “A Span Extraction Approach for Information Extraction on Visually-Rich Documents”. In: *DIL @ ICDAR Workshops*. **Best Paper Award**. 2021.
- [4] Nguyen Hong Son, **Hieu M Vu**, Tuan-Anh D Nguyen, and Minh-Tien Nguyen. “Jointly Learning Span Extraction and Sequence Labeling for Information Extraction from Business Documents”. In: *IJCNN*. **Oral**. 2022.

Education

Bachelor Degree, Computer Science (Honours Programme)

2020

UET - Vietnam National University, Hanoi

GPA: 3.83/4.00 (Rank: =1st/600+)

- **Highest Ranking Graduate.**

Honours and Awards

ML Alignment and Theory Scholars Scholarship

2025

ML Alignment and Theory Scholars (MATS) Program

A fully-funded, 4-month scholarship for conducting research on ML interpretability and alignment.

Anthropic Fellows Program for AI Safety Research

2025

Anthropic

A fully-funded, 4-month scholarship for conducting research on AI Safety.

FPT Smart Cloud Credits for Research, \$2000

2025

FPT Smart Cloud

Best Paper Award, \$1000

2021

ICDAR 2021, Workshop on Document Images and Language

Paper title: A Span Extraction Approach for Information Extraction on Visually-Rich Documents

Certificate of Highest Ranking Graduate

2020

UET - Vietnam National University

Awarded to students graduate with the highest GPA amongst the graduating class.

Top 4 Zalo AI Challenge 2018 - Voice Track (Individual participant)

2018

Zalo, VNG Corporation

Finished at 4th place on the Private Leaderboard of the Voice Gender/Accent Classification challenge.

Zalo AI Challenge is an annual Kaggle-like competition hosted by Zalo - one of the biggest tech companies in Vietnam. In 2018, the competition attracted over 700 teams competed in 3 challenges.

Experience

Research Scholar

Jan 2026 — Present

ML Alignment and Theory Scholars (MATS)

Berkeley, CA, USA

Fully funded research scholar position under the ML Alignment and Theory Scholars (MATS) program, working on interpretability and alignment of large language models.

Visiting Scholar
National University of Singapore, Advisor: Prof. Tan Nguyen
Fully funded visit to NUS to collaborate on research with Prof. Tan Nguyen.

Sept 2025 — Oct 2025
Singapore

AI Research Engineer
Torilab

July 2024 – Present
Hanoi, Vietnam

- Research on controlling LLM behaviours via activation steering and related intervention methods [1].
- LLM fine-tuning (SFT, DPO), prompt engineering, and large-scale model serving (vLLM, sglang, LitServe).
- Developed core framework for customizable chatbots with memory, tool use, and multi-turn interactions.
- Additional work: engagement detection, diffusion-based image generation, multi-speaker diarization.

AI Research Engineer
Cinnamon AI

Nov 2018 – May 2024
Hanoi, Vietnam

- Co-creator of **kotaemon** (24.7k+ ★), an open-source platform for local RAG applications.
- Developed RAG-based and LLM-powered applications for industry use cases.
- Research in Document Image Understanding: information extraction, cross-lingual pre-training, and model development.
- Published peer-reviewed papers; one Best Paper Award at DIL-ICDAR'21.
- Led data standardization initiatives and improved internal data management systems.
- Mentored engineers and contributed to technical knowledge sharing across teams.

Undergraduate Research Assistant
IOT Lab, UET - Vietnam National University

Aug 2017 — Sep 2019

- **Predictive models for wellbore data using machine learning and statistical methods.**
 - Facies/rock type classification; Time-series Analysis; Permeability Regression; Integrated Prediction Error Filter Analysis (INPEFA) curve calculation; Cumulative and Federated Learning.

Skills

Technical Fields	LLMs, ML Interpretability, Activation Steering, RAG, Agentic AI, Information Extraction, Document Understanding, NLP, Computer Vision, Image Processing, Time-series Analysis.
ML/AI Tools	Pytorch, Transformers, vLLM, SGLang, LangChain, Haystack, LlamaIndex, Tensorflow/Keras, Scikit-learn, OpenCV.
Software Development	Git, Github Action, Docker, CircleCI, DVC.
Programming Languages	Python, C/C++, Java, Shell Script.
Industrial Domains	Insurance, Manufacturing, Virtual Companion.
Environments	GCP, AWS, Linux, Windows/WSL.
Natural Languages	Vietnamese (native), English (IELTS 8.0), Japanese (JLPT N4).
Other	Problem Solving (Leetcode).

Professional Services

Reviewing

ICML 2026, ICLR 2026, MoFA @ ICML 2025, XLLM-Reason-Plan @ COLM 2025

Teaching/Mentoring

Cinnamon AI Bootcamp 2020, 2022, 2023

- Mentored groups of 3-4 students.
- Designed syllabus, prepared entrance tests, interviewed candidates.
- Prepared materials and gave lectures on Language Modelling and Transformers.

Conference/Workshop Presentations

- Angular Steering: Behavior Control via Rotation in Activation Space. *NeurIPS 2025* (Spotlight Poster); *Mechanistic Interpretability Workshop at NeurIPS 2025* (Poster).
- A Span Extraction Approach for Information Extraction on Visually-Rich Documents. *ICDAR 2021, Workshop on Document Images and Language* (Oral).

Personal Projects

Kotaemon (25.2k+ ★) - An open-source tool for local RAG application.

Jan 2024 — May 2024

Open-source project, Co-creator



- A local RAG-based tool for chatting with your documents. Built with both end users and developers in mind.
- For end users: A local Question Answering UI for RAG-based QA.
- For developers: A framework for building your own RAG-based QA pipeline.

Data Utility Improvement Experiment for DECAF

Oct 2022 — Nov 2022

Personal research



- A personal research on Causal Inference, Algorithmic Fairness and specifically the paper [DECAF: Generating Fair Synthetic Data Using Causally-Aware Generative networks](#).
- Improved data utility of the DECAF method using alternating graph during synthesis while still achieving similar level of fairness.

Gender/Accent Classification for Vietnamese short voice recordings

Aug 2018 — Sep 2018

Zalo AI Challenge 2018



- Classify the speaker's voice in a recording (typically under 3 seconds) by gender and regional accent.
- **4th place** on the Private Leaderboard, achieved **79.208% accuracy** in 10 days as an **individual participant**.

References

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